

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1507
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Reverse Engineering Task

For each task, study the described big data solution, infer how it was built, identify the underlying components, and explain what design decisions were likely made.

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy,

- encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.
4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.
 5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.

Reverse Engineering Task Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Inference Accuracy	25%	Infers system components and flow with high accuracy	Mostly accurate inference	Basic inference	Weak inference
Understanding of Architecture	25%	Shows clear understanding of underlying big data architecture	Good understanding	Partial understanding	Poor understanding
Use of Technical Evidence	20%	Uses strong reasoning and technical evidence	Reasonable evidence	Limited evidence	No meaningful evidence
Depth of Analysis	15%	Analysis is deep and insightful	Reasonably deep	Basic depth	Superficial analysis
Clarity	15%	Clear and organized explanation	Generally clear	Somewhat unclear	Poorly presented

REVERSE ENGINEERING TASK.

1. INGESTION & DISTRIBUTED STORAGE:

HIGH VELOCITY DATA INGESTION:

→ Apache Kafka / AWS Kinesis:

- Decouples producer applications from analytical customers.

→ Fluentd / Logstash: Agents collecting raw clickstream and app interaction

SCALABLE STORAGE LAYER:

→ Data Lake: Stores raw semi-structured JSON clickstreams and app interaction logs, at petabyte scale.

→ Columnar Formats: Converts raw data into Apache Parquet / ORC for efficient compression and query performance.

NOTES:-

→ To ingest billions web clicks, mobile taps, and customer transactions daily.

→ Apache Kafka → shock absorber.

→ Storage system like Amazon S3 or HDFS

19/8/2020

Reverse Engineering CLOUD COMPUTING

Data Ingestion Framework :-

A data ingestion framework collects data from different sources and transfers it into the cloud platform for storage and analysis.

Components:

(i) Data Sources:

- Databases
- IoT devices
- Applications
- Logs

(ii) Data ingestion layer

(iii) Processing layer

(iv) Storage layer

(v) Analytics layer

How data enters the platform:-

1. Website clicks
2. Mobile interactions
3. Transactions.

Reverse Engineering

✓ Batch processing

It means collecting a large amount of data and processing it together at a scheduled time. It is useful for handling large historical datasets.

Tools like

- ✓ Hadoop MapReduce
- ✓ ~~Apache~~ Spark
- ✓ Hive

These tools are commonly used.

- ✓ Hadoop divides a large data into smaller tasks.
- ✓ Spark processes data quickly in parallel
- ✓ Hive allows us to analyze the data using SQL-like queries.

DATA ORGANIZATION & INDEXING

In a big data system, we have different types of data. For example, customer details and transaction records are structured data, while JSON files, website clicks and logs are semi-structured data.

So, we organize these data separately to make processing easier.

Since the amount of data is very large, we divide it into smaller parts. This is called partitioning.

Eg: Transaction data divided based on date, customer or region.

We also use indexing, which helps us to find the required data quickly without searching the entire dataset.

Dashboard and Business Intelligence.

After the Data is collected, stored and processed, the final results need to be presented in a way that managers can easily understand.

Business Intelligence tools such as power BI, Tableau, Looker can be connected to the analytics layer.

The processed data is sent to these tools and displayed using charts, graphs and tables.

For Example: A manager can view total sales, customer activity, popular products and regional performance on single dashboard.

These dashboards help managers understand business performance and make better decisions based on the data.

- Instead of manually analyzing billions of records, managers can simply look at the important information presented by the dashboard.

Therefore, the BI layer converts complex analytical results into simple and useful information for decision-making.

Scalability and Partitioning

When the amount of data and number of users increase, the analytics platform must be able to handle the additional workload.

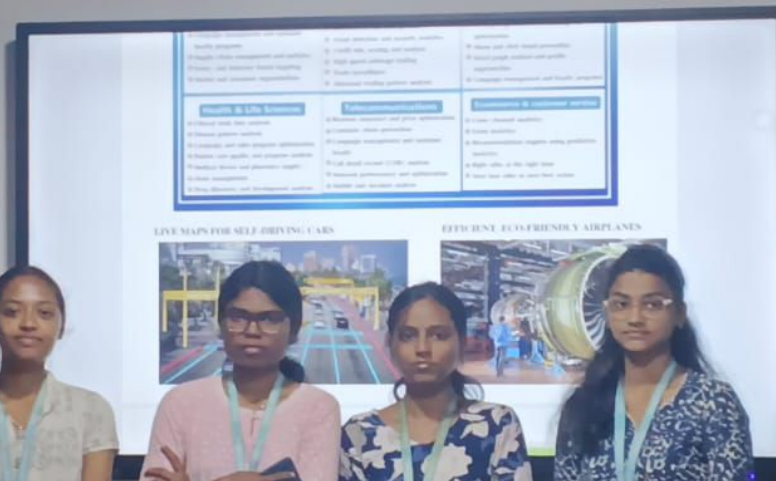
One important approach is horizontal scaling. Instead of continuously upgrading one powerful machine, more machines can be added to the system.

The data can also be divided using partitioning - for example based on date, customer, region, etc.

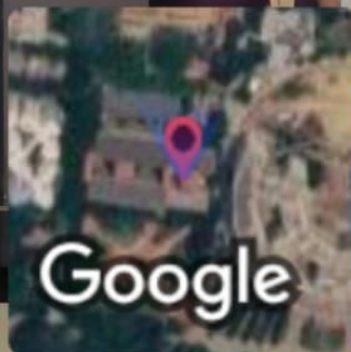
The system can also use replication to maintain multiple copies of data and load balancing to distribute requests among different machines.

These techniques improve the system's performance, availability, fault tolerance.

For eg: if one machine fails, another machine containing a replicated copy can continue serving the data.



GPS Map Camera



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